Java – Else-If Ladder: Logic Building Questions

Practice these using if, else-if, else and logical conditions.

1. Electricity Bill

Write a Java program to calculate an electricity bill based on units consumed:
• First 100 units → ■2/unit
• 101–200 units → ■3/unit
• 201–400 units → ■5/unit
• Above 400 units → ■7/unit
If the bill exceeds ■2000, add a 5% surcharge.
Example: Input 450 units → calculate the bill and surcharge if applicable.

2. Student Result & Grade

Take marks for 5 subjects. Calculate total and percentage. If any subject is below 33, print **Fail**. Otherwise assign a grade:
90%+ → A+ | 80–89 → A | 70–79 → B | 60–69 → C | 50–59 → D | Below 50 → E.

3. Triangle Classification

Take three sides **a, b, c**. First check whether they can form a triangle using:
a + b > c, b + c > a, and c + a > b.
If valid: all three equal → Equilateral; any two equal → Isosceles; all different → Scalene. Otherwise print **Invalid Triangle**.

4. Salary Tax Calculator

Take an employee's annual salary and calculate tax using these slabs:
■0–■3,00,000 → No tax
■3,00,001–■6,00,000 → 5%
■6,00,001–■10,00,000 → 10%
■10,00,001–■15,00,000 → 20%
Above ■15,00,000 → 30%
Display salary, tax, and salary after tax. **Challenge:** handle the slabs correctly.

5. ATM Withdrawal

Take **balance, withdrawal amount, and PIN**. Apply these rules:
1) Incorrect PIN → Invalid PIN
2) Correct PIN → withdrawal must be greater than 0
3) Amount must be a multiple of ■100
4) Amount cannot exceed balance
5) Maximum withdrawal per transaction is ■20,000
If everything is valid, deduct the amount and display the remaining balance.

Tip: In an else-if ladder, conditions are checked from top to bottom. Once one condition is true, its block runs and the remaining else-if conditions are skipped.